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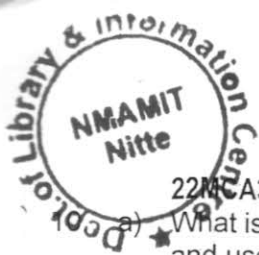
22MCA351 – CLOUD COMPUTING AND BIG DATA ANALYTICS

Max. Marks: 100

Unit – I

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) Explain unstructured data, how to deal with it, and the techniques used to find patterns in unstructured data.	10	L* 2	1	1
	b) Explain the difference between Traditional Business Intelligence (BI) and Big Data.	4	L2	1	1
	c) Describe the features and challenges of Big Data.	6	L2	1	1
2.	a) Describe a typical Data Warehouse environment and its components.	8	L2	1	
	b) Differentiate between structured and semi-structured data, highlighting their characteristics and examples.	6	L2	1	1
	c) Discuss the applications of Big Data across various industries.	6	L2	1	1
Unit – II					
3.	a) Describe the Shared Nothing Architecture and discuss its advantages and disadvantages.	8	L2	2	1
	b) Explain the various categories of data analytics and illustrate how these progress in terms of difficulty and business value.	6	L2	2	1
	c) Explain the difference between Parallel Systems and Distributed Systems.	6	L2	2	1
4.	a) Discuss the CAP Theorem and its significance in distributed systems.	10	L2	2	1
	b) List and define the key terminologies used in Big Data environments.	10	L2	2	1
Unit – III					
5.	a) Describe the YARN architecture and its components.	10	L2	3	1
	b) Explain Hadoop, highlighting its key features and advantages.	10	L2	3	1
6.	a) Explain data processing in Hadoop, including MapReduce phases and daemons.	6	L2	3	1
	b) Differentiate between Hadoop and SQL.	6	L1	3	1
	c) Explain the different versions of Hadoop and their features.	8	L2	3	1
Unit – IV					
7.	a) Discuss various cloud deployment models and their characteristics.	10	L2	4	1
	b) Explain the factors to be considered when opting for Public Cloud and compare the different Public Cloud service models.	10	L2	4	6
8.	a) Explain the different cloud service models with suitable examples.	10	L2	4	1
	b) Describe the key components of private cloud building, including Physical Layer, Virtualization Layer, and Cloud Management Layer.	10	L2	4	1
Unit – V					
9.	a) Discuss the advantages and challenges of adopting cloud solutions in business operations.	10	L2	5	1
	b) Explain application, server, and network security in cloud computing with examples.	10	L2	5	1

PTQ



22MCA351

SEE – December 2025

- a) What is RightScale Cloud Management System? Explain its Key features and uses.
- b) What is multi-cloud management? Explain the features and list out the benefits.

10	L1	5	1
10	L1	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.

UNIT - I: DATA ANALYTICS

Max. Marks: 100

BT* CO* PO*

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P.T.O.